EV BATTERY INTELLIGENCE SYSTEM

AI-Powered Lithium-Ion Battery State of Charge Estimation

Technical Evaluation Report | Generated: 06 November 2026

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EXECUTIVE SUMMARY

This report presents an AI-powered battery State of Charge (SOC) estimation system developed using three recurrent deep-learning architectures: Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Bidirectional LSTM (BiLSTM). The models were trained on prepared lithium-ion battery time-series data and evaluated using validation and independent test datasets.

The independent test evaluation covers four temperature conditions: -10°C, 0°C, 10°C and 25°C. The following sections summarize the model configurations, evaluation metrics, prediction analysis and key findings.

1. SYSTEM OVERVIEW

Project Parameter	Description
Dataset	LG 18650HG2 Lithium-Ion Battery Dataset
Training Observations	669,956
Input Features	5 prepared input features
Sequence Length	30 time steps
Sequence Stride	5
Prediction Target	Battery State of Charge (SOC)
Models	LSTM, GRU and BiLSTM
Test Temperatures	-10°C, 0°C, 10°C and 25°C

2. MODEL ARCHITECTURES

Model	Recurrent Configuration	Additional Layers
LSTM	LSTM (64 units)	Dropout (0.2), Dense (32), Output (1)

Model	Recurrent Configuration	Additional Layers
GRU	GRU (64 units)	Dropout (0.2), Dense (32), Output (1)
BiLSTM	Bidirectional LSTM (64 units per direction)	Dropout (0.2), Dense (32), Output (1)

The models use the Adam optimizer and Mean Squared Error loss. Training used a maximum of five epochs with early stopping patience of two epochs.

3. VALIDATION PERFORMANCE

Model	MAE	RMSE	R2
LSTM	0.032721456	0.04749193	0.9736713
GRU	0.051092524	0.061486296	0.9558687
BiLSTM	0.023063626	0.036227863	0.9846794

4. INDEPENDENT TEST EVALUATION

The trained models were evaluated on separate test files representing four battery temperature conditions.

Overall Model Performance

Model	MAE	RMSE	R2
BiLSTM	0.0184181	0.024350975	0.99020547
LSTM	0.024953406	0.032144595	0.9834721
GRU	0.039756697	0.047107656	0.9678184

Temperature-wise Performance

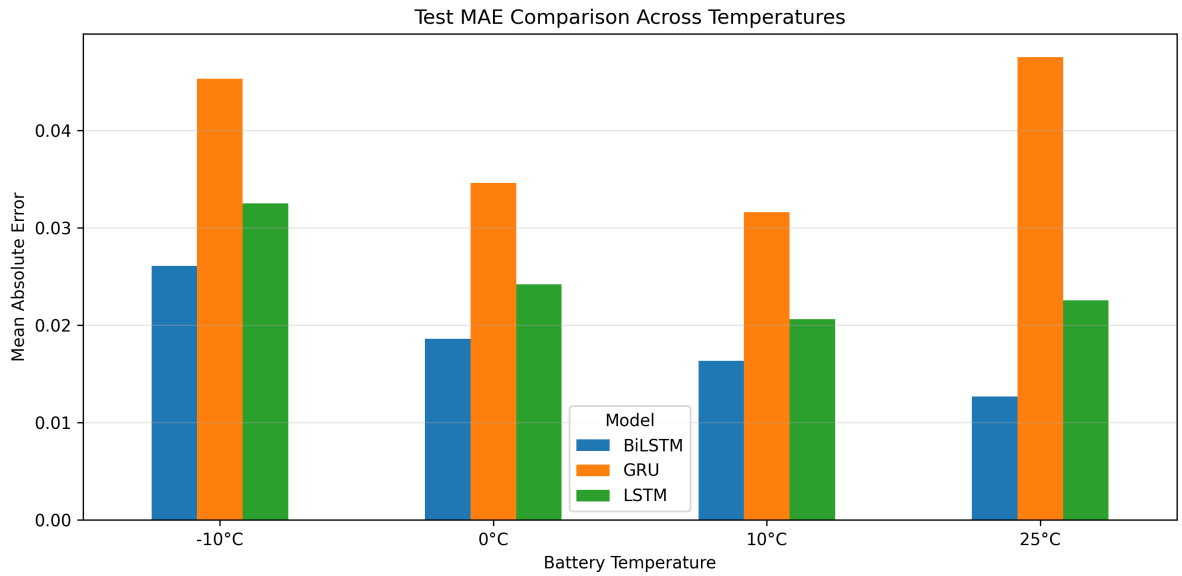
Temperature	Model	MAE	RMSE	R2	Observations
-10°C	LSTM	0.03250599	0.047610115	0.96289504	7853
-10°C	GRU	0.045301966	0.057662107	0.945573	7853
-10°C	BiLSTM	0.02606983	0.037280697	0.9772489	7853
0°C	LSTM	0.024177002	0.030863948	0.9867354	8501
0°C	GRU	0.03460613	0.0422379	0.9751575	8501
0°C	BiLSTM	0.01857955	0.024145251	0.9918819	8501
10°C	LSTM	0.020605389	0.023991233	0.9923577	8851
10°C	GRU	0.031599585	0.03824632	0.9805777	8851
10°C	BiLSTM	0.016350185	0.019762112	0.9948145	8851

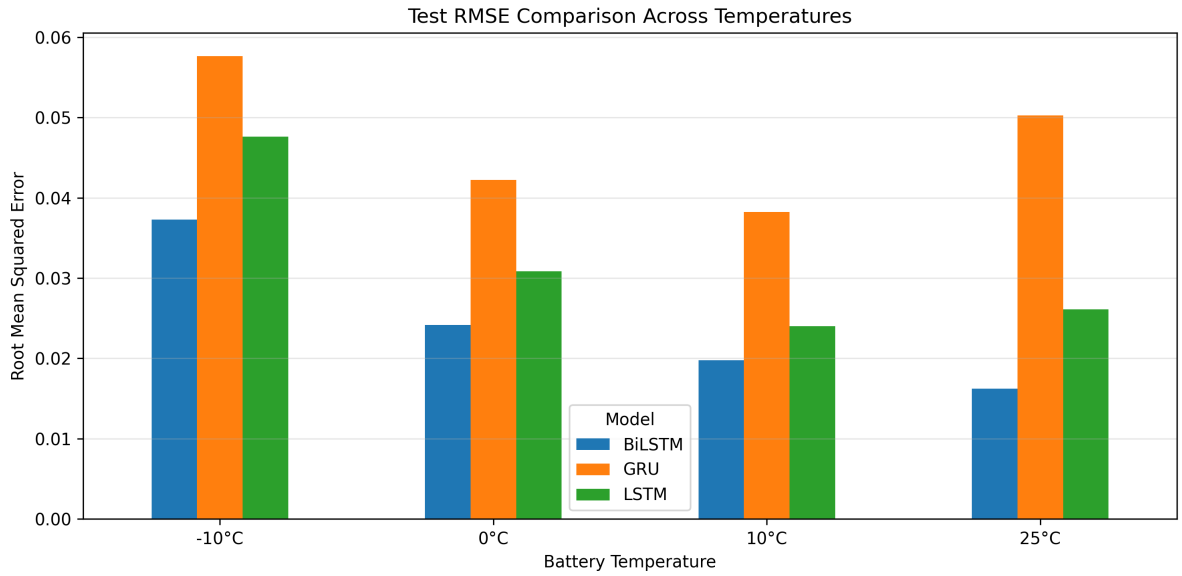
Temperature	Model	MAE	RMSE	R2	Observations
25°C	LSTM	0.022525247	0.026113082	0.9919002	9498
25°C	GRU	0.047519106	0.050284296	0.9699653	9498
25°C	BiLSTM	0.01267283	0.016215838	0.99687654	9498

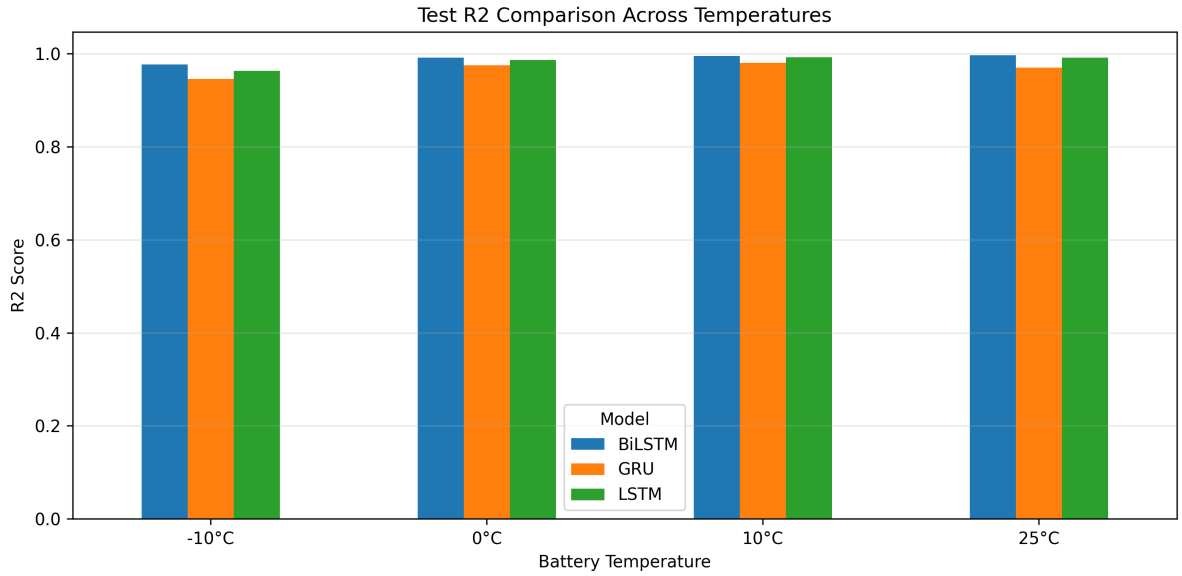
5. SOC PREDICTION ANALYSIS

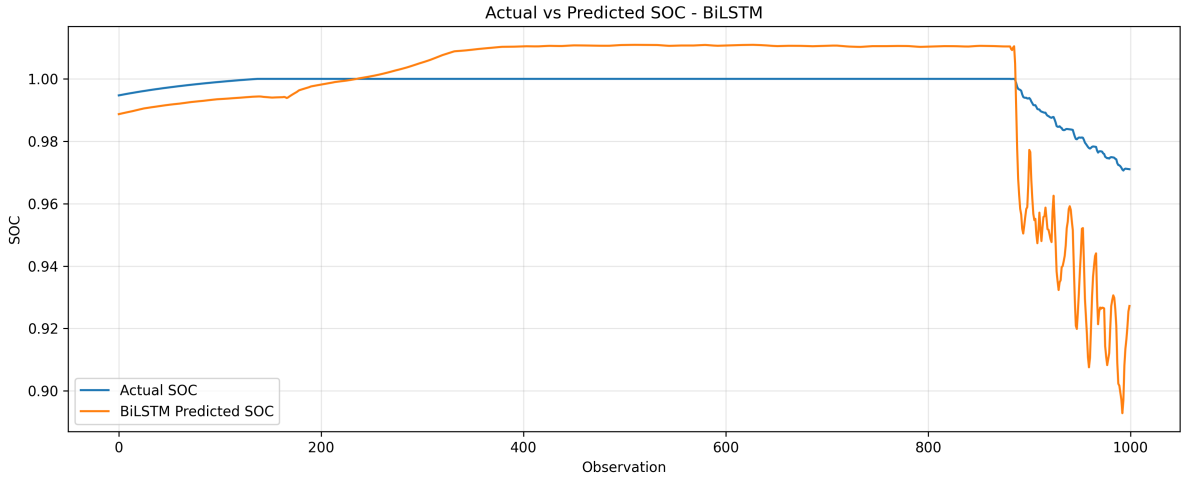
Prediction Metric	Value
Total Predictions	34,703
MAE	0.01809
RMSE	0.02508
R^2 Score	0.99153

6. PERFORMANCE VISUALIZATIONS









7. KEY FINDINGS

The independent test results indicate that BiLSTM achieved the strongest overall performance among the three evaluated architectures, based on the reported average test MAE, RMSE and R^2 metrics. Temperature-specific results should be

considered when interpreting model performance.

8. LIMITATIONS AND SCOPE

This project evaluates prepared experimental battery datasets. The dashboard analyzes saved model predictions and is not connected to a live electric vehicle battery. The reported results apply to the dataset and test conditions used in this experiment; they do not by themselves establish performance on every battery chemistry, vehicle or operating environment.

9. CONCLUSION

The EV Battery Intelligence System demonstrates the use of recurrent deep-learning architectures for lithium-ion battery SOC estimation. By comparing LSTM, GRU and BiLSTM models and evaluating their predictions under multiple temperature conditions, the project provides a structured basis for analyzing model performance and prediction errors.